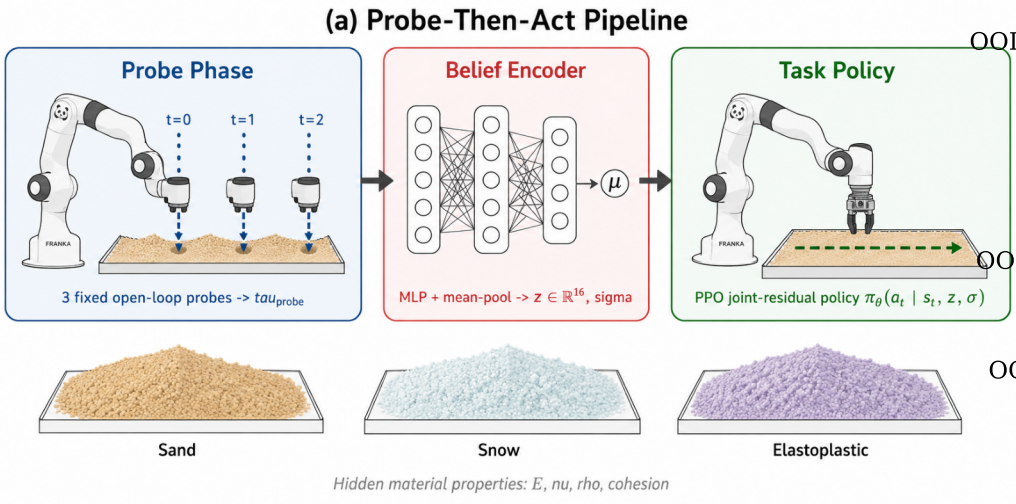
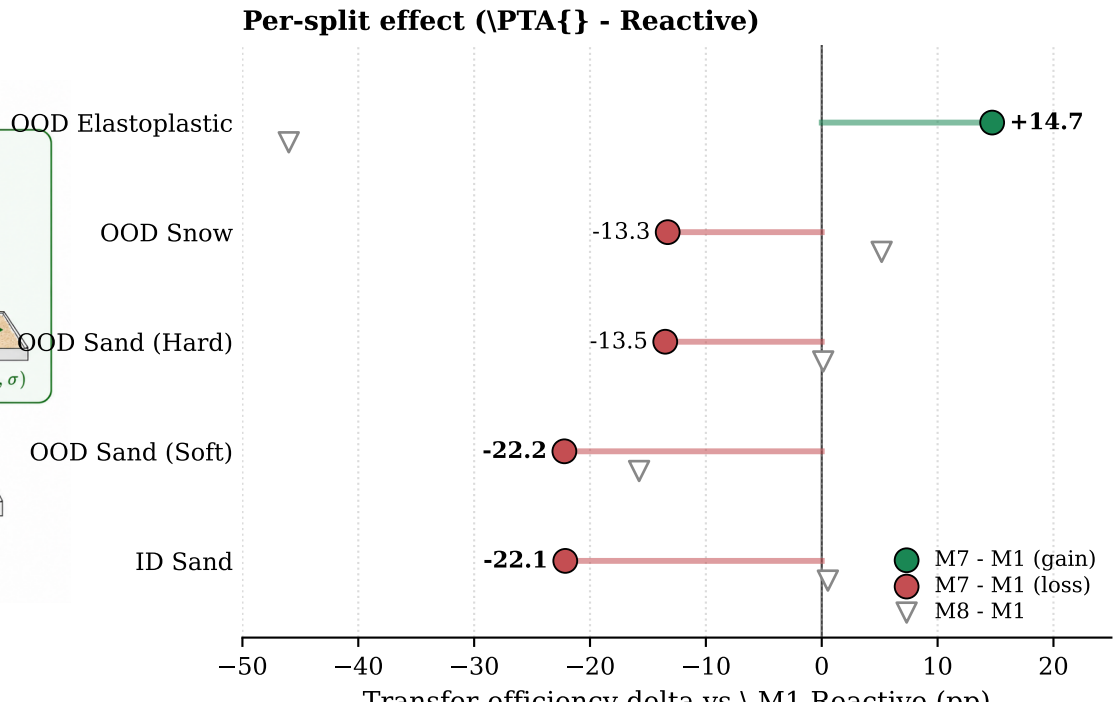

(a) Probe-Then-Act Pipeline
Probe Phase
t=0
t=1
t=2
3 fixed open-loop probes -> tau_{probe}
Belief Encoder
μ
MLP + mean-pool -> $z \in \mathbb{R}^{16}$, sigma
Task Policy
PPO joint-residual policy $\pi_\theta(a_t \mid s_t, z, \sigma)$
Sand
Snow
Elastoplastic
Hidden material properties: E, nu, rho, cohesion


Per-split effect (\PTA{} - Reactive)
OOD Elastoplastic
OOD Snow
OOD Sand (Hard)
OOD Sand (Soft)
ID Sand
+14.7
-13.3
-13.5
-22.2
-22.1
M7 - M1 (gain)
M7 - M1 (loss)
M8 - M1
−50
−40
−30
−20
−10
0
10
20
Transfer-efficiency delta vs.\ M1 Reactive (pp)